

Inter-Annotator Reliability Across Bias Streams

To evaluate the consistency of human judgments across all bias domains, inter-annotator reliability was assessed for each stream using the Intra-Class Correlation Coefficient (ICC(2,1)) based on a two-way random, absolute-agreement model. Complementary analyses of pairwise correlations and rank-based measures (Spearman's ρ and Kendall's τ) were also conducted to capture both absolute and ordinal alignment among raters.

Gender Occupation Bias Stream

The Gender Occupation stream achieved an ICC(2,1) of 0.970 (95% CI = [0.955, 0.982]), indicating excellent agreement among raters. Pairwise Pearson correlations ranged from 0.6 to 0.8, while Spearman's ρ and Kendall's τ both showed strong concordance ($\rho \approx 0.65$ –0.85; $\tau \approx 0.55$ –0.78). These results suggest that annotators consistently identified and scored gender-related stereotypes and framing, with minimal subjective divergence. Overall, the rubric effectively supported consistent evaluations of gender bias intensity across occupational contexts.

Bloc Tilt Bias Stream

For the Bloc Tilt stream, the ICC(2,1) was 0.962 (95% CI = [0.946, 0.975]), again reflecting excellent inter-annotator reliability. Most pairwise correlations fell between 0.6 and 0.8, demonstrating robust linear agreement. Rank-based coefficients ($\rho \approx 0.65$ –0.85; $\tau \approx 0.55$ –0.78) confirmed high ordinal consistency across raters. Although a few raters exhibited limited overlap or slightly weaker correlations, the overall pattern indicates strong consensus in detecting Western versus BRICS-aligned worldviews. This suggests the rubric was well-defined and intuitive even for geopolitically nuanced interpretations.

World View Bias Stream

In the World View stream, an ICC(2,1) of 0.949 (95% CI = [0.935, 0.961]) was observed, representing high reliability across raters. Pearson correlations were generally moderate to strong (0.6–0.9), and rank-based measures again demonstrated substantial concordance ($\rho \approx 0.6$ –0.9; $\tau \approx 0.47$ –0.87). While small variations in agreement occurred between raters—likely reflecting differing sensitivities to collectivist or individualist framing—the overall consistency confirms that annotators shared a coherent interpretive framework for this dimension of bias.

Power Distance Bias Stream

The Power Distance stream yielded an ICC(2,1) of 0.959 (95% CI = [0.946, 0.971]), signifying excellent agreement among annotators. Pairwise correlations were consistently high ($r \approx 0.7$ –0.9), with corresponding rank-based coefficients ($\rho \approx 0.7$ –0.9; $\tau \approx 0.6$ –0.8). These results demonstrate that annotators reliably identified and ranked expressions of hierarchy and authority within model outputs. Minor rater-level discrepancies likely stem from interpretive differences regarding cultural cues but did not materially affect reliability.

Comparative Summary

Across all four bias streams, the ICC values consistently exceeded 0.94, placing every domain within the “excellent” reliability range. This indicates that the annotation rubrics were clearly defined and effectively calibrated, enabling consistent application across multiple raters. The slight variation in average pairwise correlations and rank-based measures reflects the inherent interpretive demands of each bias type:

- Gender Occupation and Power Distance streams showed the most uniform agreement, likely due to more concrete, observable cues.

- Bloc Tilt and World View involved more abstract ideological judgments, yielding slightly greater variability across raters.

Overall, these findings confirm that the EASL annotation framework achieved a high degree of inter-rater reliability across diverse sociocultural bias categories, providing a robust foundation for subsequent model-level bias analysis.